

```
gatelevel.v
~/Desktop/exp5

1 module boolean_gatelevel(input a,b,c,d,output f);
2 wire na,nb,nc,nd,t1,t2,t3,t4,t5,t6;
3 not(na,a);
4 not(nb,b);
5 not(nc,c);
6 not(nd,d);
7
8 and(t1,na,nb,nc,d);
9 and(t2,na,b,nc,nd);
10 and(t3,na,b,c,nd);
11 and(t4,a,nb,nc,d);
12 and(t5,a,nb,c,nd);
13 and(t6,a,b,nc,nd);
14 or(f,t1,t2,t3,t4,t5,t6);
15
16 endmodule
```

```
logical.v
~/Desktop/exp5

1 module boolean_logical(input a,b,c,d,output f_sop,f_pos);
2
3 assign f_sop = (~a&b&c&d)|(~a&b&~c&d)|(~a&b&c&~d)|(a&~b&c&d)|(a&b&~c&d);
4
5 assign f_pos = (a|b|c|d)& (a|b|~c|d)& (a|b|~c|~d)& (a|~b|c|~d)& (a|~b|~c|~d)& (~a|b|c|d)& (~a|b|~c|~d)& (~a|~b|c|d)& (~a|~b|~c|d);
6 endmodule
```

```
conditional.v
~/Desktop/exp5

1 module boolean_conditional(input a,b,c,d,output f);
2 assign f = (!a && !b && !c && d)?1'b1:
3           (!a && b && !c && !d)?1'b1:
4           (!a && b && c && !d)?1'b1:
5           (a && !b && !c && d)?1'b1:
6           (a && !b && c && !d)?1'b1:
7           (a && b && !c && !d)?1'b1: 1'b0;
8 endmodule
```

```
test.v
~/Desktop/exp5

1 module test;
2 reg a,b,c,d;
3 wire f_sop,f_pos,f_con,f_gates;
4 boolean_logical bl(.a(a),.b(b),.c(c),.d(d),.f_sop(f_sop),.f_pos(f_pos));
5 boolean_conditional bc(.a(a),.b(b),.c(c),.d(d),.f(f_con));
6 boolean_gatelevel bg(.a(a),.b(b),.c(c),.d(d),.f(f_gates));
7
8 initial begin
9 $dumpfile("test.vcd");
10 $dumpvars(0,test);
11
12 $display("a b c d | f_sop f_pos f_con f_gates");
13 $display(".....");
14
15 for(integer i=0;i<10;i=i+1) begin
16 {a,b,c,d}=i;
17 #1;
18
19 $display("%b %b %b %b | %b %b %b %b",a,b,c,d,f_sop,f_pos,f_con,f_gates);
20 end
21 $finish;
22 end
23 endmodule
```

```

cseeb1@mits-H610M-S2-DDR4:~/Desktop/exp$ gedit logical.v
cseeb1@mits-H610M-S2-DDR4:~/Desktop/exp$ gedit conditional.v
cseeb1@mits-H610M-S2-DDR4:~/Desktop/exp$ gedit gatelevel.v
cseeb1@mits-H610M-S2-DDR4:~/Desktop/exp$ gedit test.v
cseeb1@mits-H610M-S2-DDR4:~/Desktop/exp$ iverilog -o result logical.v conditional.v gatelevel.v test.v
cseeb1@mits-H610M-S2-DDR4:~/Desktop/exp$ vvp result
VCD info: dumpfile test.vcd opened for output.
a b c d | f_sop f_pos f_con f_gates
.....
0 0 0 0 | 0 0 0 0
0 0 0 1 | 1 1 1 1
0 0 1 0 | 0 0 0 0
0 0 1 1 | 0 0 0 0
0 1 0 0 | 1 1 1 1
0 1 0 1 | 0 0 0 0
0 1 1 0 | 1 1 1 1
0 1 1 1 | 0 0 0 0
1 0 0 0 | 1 0 0 0
1 0 0 1 | 1 1 1 1
1 0 1 0 | 0 1 1 1
1 0 1 1 | 0 0 0 0
1 1 0 0 | 1 1 1 1
1 1 0 1 | 0 0 0 0
1 1 1 0 | 0 0 0 0
1 1 1 1 | 0 0 0 0
cseeb1@mits-H610M-S2-DDR4:~/Desktop/exp$ gtkwave test.vcd

```

